ASSIGNMENT 1

Unit Covered	Unit I — Image Formation & Representation, Image Processing
Suggested Tools	Python (NumPy, OpenCV, Matplotlib) or MATLAB

1. Take (or download) three images of the same well-lit indoor scene: one normal, one deliberately under-exposed, and one over-exposed. Explain, using the radiometry and digitization concepts from class, why the same scene produces such different pixel value distributions.
2. Simulate the CCTV's "grid artefact" complaint by writing your own 2D convolution function (no library shortcuts) and running a low-pass (blur) and a high-pass (sharpen) kernel over one of your images. Compare your output against OpenCV's built-in filter2D to prove correctness.
3. Take the histogram of your under-exposed image, apply histogram equalization, and write a short "incident report" (200–250 words) explaining, in a client-friendly tone, why equalization improved visibility and what its limitations are.
4. Compute the 2D Fourier transform of one image, visualise the magnitude spectrum, and use frequency-domain filtering to remove the suspected grid pattern (simulate it by adding a synthetic periodic noise pattern to a clean image first, then remove it).
5. Finally, apply a rigid transformation (rotation + translation) and an affine transformation to one image and explain, geometrically, which one a camera undergoing pure panning would produce versus one mounted on a slightly warped bracket.